HAW Kiel
Social Media Analytics
Capstone Project

Development of a combined Graph-RAG System to Analyze and Compare the Television Adaption of Pretty Little Liars with its book source.

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1 Introduction

In today's day and age, AI tools have quickly taken over the market in many different applications. The perhaps most prominent AI-solutions are so called Large Language Models (LLMS), with popularly used instances such as ChatGPT, Gemini or Claude.

While LLMs have shown great results for a huge variety of problems, they have a tendency to hallucinate information, thus making them an unreliable tool (Gao et al., 2023). There are a few different solutions to such issues, with the most influential solution being so called Retrieval-Augmented Generation (RAG) systems.

RAG systems combine information retrieval with a generative AI model, which reduces hallucinations (Lewis et al., 2020, p. 6). That being said, so called RAG-Systems still offer a few limitations and issues. Often such systems struggle with understanding the bigger context, for example often not understanding sarcasm or parodies, taking the information as true (Williams, 2024). As a more robust solution, the RAG System can be combined with a graph based architecture, where knowledge is sorted into so called "entity-relationship-graphs", which increases stability across multiple documents (Dimitrova, 2025, p. 40).

In the following experiment such a combined approach of a RAG system with a graph database will be implemented and tested in the comparison of the plots of a series adaption with its original book source. This way we can better analyse its ability in understanding relationships and contexts across multiple documents.

2 Methodology

2.1 Planned Architecture

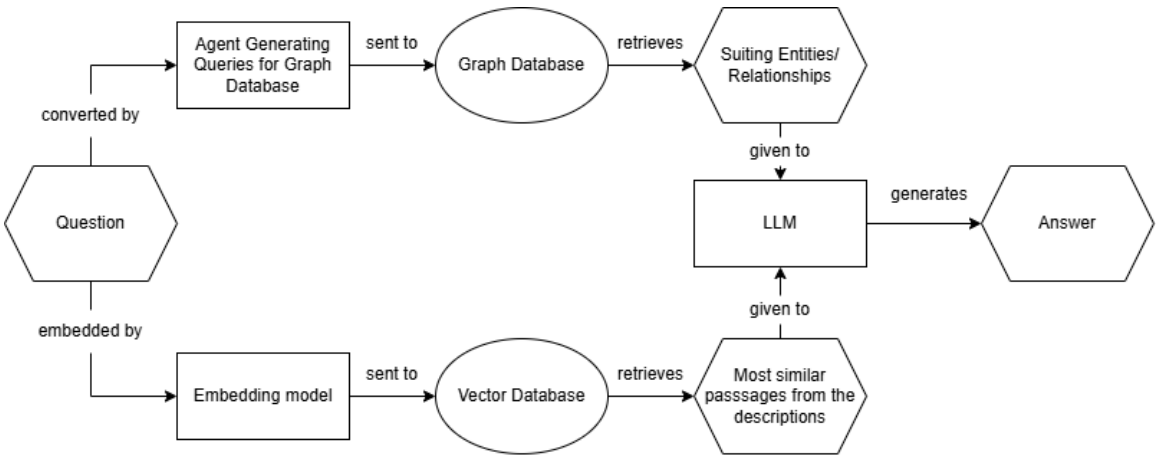


Figure 1: Architecture the implemented Graph-RAG-System

The proposed architecture, as demonstrated in Figure 2, combines a regular RAG system with a so called GraphRAG System.

2.1.1 Regular RAG system

The RAG system in this experiment is made up of two main components: A vector database and an embedding model. As an embedding model “harrier-oss-v1-270m” by Microsoft was used as it offers a very good balance of model size and embedding quality (“MTEB Leaderboard - a Hugging Face Space by mteb,” n.d.). The summaries were then separated into individual chunks of a length of around 500 words as previous studies deem this value to provide the highest faithfulness of responses (Wang et al., 2024, p. 5). Each chunk was individually embedded into a vector which make up the information inside the vector database. Upon entering a question, the model once again embeds the question into a vector and then retrieves the most similar vectors, who are hopefully therefore also the most similar passages to the question, and gives them to the LLM as context.

In order to increase performance without reducing too much speed, a reranking architecture was also implemented: The first embedding model, usually of smaller model size, takes a larger than normally used number of passages from the text. Then a larger more powerful but also slower model is used to rerank the previously selected passages in similarity to the given input text. This way the performance of the retrieval can be increased without having to compromise on speed of the algorithm.(Zhang et al., 2025, p. 1) In this example “Qwen3-Reranker-0.6B” was used as the reranking algorithm as it proved to be a good balance between performance and speed in previous tests.

2.1.2 GraphRAG System

The basis of the GraphRAG system is in its Graph Database. A graph database contains entity relationship graphs, which are made up of different entities in the texts (so called nodes) and their respective relationships towards one another (known as edges). Given a question, the retrieval system then selects the relevant nodes and their edges and returns them to the LLM as context.

To reduce room for errors unrestricted code generation will not be allowed. Instead, restricted tooling is implemented, where the LLM is able to select from a small set of predefined graph query functions. The LLM is allowed choose from three different functions: 1. Retrieving all relationships (and their respective nodes) from one given node. 2. Retrieving all relationships between two nodes and 3. Retrieving all relationships and their respective nodes from one node for a given relationship type.

By choosing one of these functions and the suiting parameters, the agent then can select the most relevant information from the identity-relationship graph, which is then given to the final LLM as context to answer the given question. This approach can be implemented without any prior training (Sun et al., 2023, p. 4) and was found to be more stable in prior experiments.

2.2 Evaluation of results

As data the fictional story of the popular TV series Pretty Little Liars is being used. Pretty Little Liars first aired in the year 2010 ("Pretty Little Liars," 2012) and centres four friends who lose touch after their fifth friend, Alison DiLaurentis, mysteriously disappears. A year after her disappearance, the four main characters all start to receive threatening text messages by an anonymous entity, otherwise known as "A". The seven seasons centre around answering two key questions: 1. Who is A and 2. What happened to Alison? (Bingham, 2014, p. 96 f.) This data is believed to be especially fitting as there are not only many re-appearing characters in both the book and the series, but also because the series adaption is quite different in its characters and relationships compared to its source.

To avoid copyright issues and keep the data small enough for the scope of this project, this project will only be focusing on summaries provided by the website prettylittleliars.fandom.com. Fandom.com is a large wiki platform maintained by fan communities about different universes from movies, TV shows, video games etc. and offers various content including plot summaries of the respective medium ("What Is Fandom," n.d.).

As a baseline a test sed was created, made up of five questions about the material, with optimal answers, acquired by research on the mentioned wiki platform. These answers can then be used to compare the generated response. While the structure of the answer is irrelevant, the focus will lay on the content of the answer, evaluating its faithfulness to the given material.

The selected questions and their correct answers are the following:

Question	Answer
Who is „A"?	In both the books and the series, A is revealed to be Mona Vanderwaal.

Who is Caleb?	In the series Caleb is homeless and secretly lives in the school. Later he becomes Hanna's boyfriend. In the books he does not exist.
Who did Spencer date?	In the books Spencers boyfriend is Wren, while in the series Spencer first dates Alex Santiago and then briefly Wren and later Toby Cavanaugh.
What happens to Ezra Fitz?	In the books, Ezra Fitz is the English teacher and has a brief relationship with Aria, before being arrested. In the series he is also her English teacher and has a relationship with her but is never arrested for it.
What is the relationship between Melissa and Spencer?	Melissa is Spencers sister in both the books and the series.
What is the relationship between Hanna and Lucas?	While in the series Lucas and Hanna are just friends, in the books they have a brief romance.

Table 1: Test questions and their correct answers.

2.3 Data Collection

Fandom.com is built on the MediaWiki software, which provides a freely accessible custom API that was utilized for the implemented web scraping algorithm ("Sandbox/API," n.d.). However, the API appears to be inconsistent across individual fandoms. When accessing a page of the Pretty Little Liars subsection via the API, the response typically consists of the full HTML document. Additionally, these documents often lack consistent use of identifiers or classes and do not follow a uniform structural pattern. This inconsistency significantly complicated the automated extraction of relevant data and slowed the development process. While the implemented scraping algorithm functions reliably for the specific dataset used in this study, it remains relatively fragile and is not very likely to generalize well to other pages or subwikis.

Due to the complexity of scraping the data and the limited scope of this project, the focus will only be on the first two seasons of the show and the first four books in the series, as at the end of the second season or the fourth book, the first story arc is closed for each medium by revealing the identity of the first instance of “A” (“Pretty Little Liars (Book series),” n.d.). This decision was made to keep the scope of the project appropriate, while still being able to retain enough information to quiz the system on more complicated questions.

Unfortunately, focusing only on fan-written summaries does offer a few limitations: While the summaries per episode are rather detailed, in the context of the books, the website offers only summaries per book, not per chapter, rendering those summaries more general, skipping a few of the less important details. These details are therefore also missing in the knowledge bases. Further, the summary of the third book “Perfect” is completely missing. Therefore, it had to be replaced by another summary taken from a reddit post (highlycaffinatedveg, 2023).

2.4 Preprocessing

To use the data for the regular RAG system, no further preprocessing was needed.

This was not the case for the Graph system. As the scraped data is in a textual format, while the graph database expects a graph-based-format, the entities and relationships needed to be extracted from the given text. At first a local LLM was implemented for this purpose, but unfortunately due to limited computing power and thus the limited sizes of possible LLMs to be used, the data was often incorrect, both in its content as well in its format. The different LLMs that were tried (including but not limited to qwen3:4b, Zephyr-7B-beta and llama3.1:8b) all struggled in following the given instructions. Often inventing new relationship types, characters or greatly misunderstanding the actual relationships given in a passage. The only solution was a bigger model, which was then realised through AIStudio using gemini-3.1-flash-lite.

Another factor in the quality of the extracted data was the used prompt. Generally more clear and strict prompts with very strict rules seemed to work best for this approach, for example defining the available categories for entities and relationship types. To keep the scope of the project from being too large, the focus on two entity types: Individual Characters and Groups. For the relationship types, the LLM was allowed to choose between the following: "ENGAGED", "FRIEND", "FAMILY", "SIBLING", "STEP-SIBLING", "PARENT", "CHILD", "DATED", "MARRIED", "BLACKMAILED", "ALLIED".

As text, the LLM was given one summary at a time, either from one episode or one book, as a whole to reduce redundancy.

After that, the following steps were implemented to clean the data and improve consistency:

1. Normalizing the names (e.g replacing just the first name by the first and last name) to not create redundant nodes.
2. Grouping different instances of one character into one instance to avoid redundant nodes
1. Normalizing relationship types as the LLM had a tendency to slightly alter them (eg. changing “FRIEND_OF” to “FRIEND”)
2. Grouping the same relationships (meaning same target, source and relationship type) were together unless another relationship type of the same source and target was in between. That was done to also consider possible changes in relationships over time (eg. going from being friends, to enemies, back to friends).

3 Results

These are the general results generated by the System:

Question	Correct Context	Generated Answer
Who is „A“?	Yes, except for series vector data.	**Books**: The identity of "A" is revealed to be Mona. **Series**: The provided context does not explicitly state the identity of "A," only that the girls are constantly searching for the person's identity and that "A" is a black-gloved figure who blackmails the Liars and Dr. Sullivan.
Who is Caleb?	Yes, but only party for series vector data.	**Books**: There is no mention of a character named Caleb. **Series**: Caleb Rivers is a character who has a relationship with Hanna Marin (dating and engaged). He is involved in shady business deals, has a foster mother named Janet, and has interactions with characters such as Jenna, Garrett, and Darren.

Who did Spencer date?	Yes except for series graph data and series vector data.	**Books**: Wren Kim:Spencer dates her sister Melissa's ex-boyfriend. **Series**: Wren Kingston: Dated in season 1. Alex Santiago: Dated in season 1. Ian Thomas: Dated in season 1. Toby Cavanaugh: Dated in seasons 1 and 2.
What happens to Ezra Fitz?	Yes.	**Books**: Ezra dates Aria Montgomery. Eventually, "A" sends pictures of the two of them to Sean, who takes the evidence to the police. As a result, the police come to Ezra's house and arrest him. **Series**: Ezra dates Aria Montgomery and deals with the consequences of their relationship. He considers resigning from his teaching position, leaves school to find a new job to make their relationship work, and reconciles with Aria. He is also involved in a situation where he is blackmailed by Noel.
What is the relationship between Melissa and Spencer?	Yes.	**Books:** Melissa and Spencer are siblings who share a volatile and competitive relationship. They frequently clash, with incidents including Spencer pushing Melissa down the stairs in a fit of rage and Spencer suspecting that Melissa killed Alison due to jealousy over Ian. **Series:** Melissa and Spencer are sisters who maintain a strained and often argumentative relationship. Their interactions are marked by tension, suspicion, and conflict—often involving their shared romantic interests (Wren and

		Ian)—though they occasionally attempt to maintain civility and share personal secrets, such as Melissa's pregnancy.
What is the relationship between Hanna and Lucas?	Yes.	**Book:** Hanna and Lucas are friends. They spend time together, including going on a "non-date" in a hot air balloon where they share personal secrets and kiss. Lucas also helps Hanna by saving her from humiliation and providing her with clothes after she is bullied. **Series:** Hanna and Lucas are friends. Their dynamic involves Hanna occasionally acting as a wingman for Lucas, such as helping him secure a date with a girl named Danielle. Lucas is sometimes suspicious of Hanna's intentions, but he is ultimately grateful for her help. Additionally, Hanna has shown empathy toward Lucas, noting that he reminds her of her own "dork days," though this sometimes causes friction in her relationship with Sean.

Table 2: Generated Results, Red = False Information

As shown in the table, the results of the combined approach seem very promising. While the agent does sometimes show issues in selecting the correct context, especially for vector-based data, it mainly generates correct answers, with false information only being in two instances of the six responses.

For the first falsely answered question of "Who is "A"?", the second part of the answer was fully incorrect. In fact, it explicitly states in the summary of the last episode "*The girls realize that Mona is in fact 'A'*.". Although relationship of Mona being "A" is not reflected in the graph database, due to the algorithm being restricted to only certain relationship types, none of which could reflect that connection. Therefore, it is likely that without explicitly clarified relationship types, the agent would have been able to answer this question.

Most of the response to the second falsely answered question is in fact correct, the only false information being, Spencer and Ian having dated. When looking at the generated context, it becomes apparent that this misbelief stems from the graph database, where Spencer and Ian's relation is classified as "DATED" with the given context being "*Spencer suggests Ian may have been into younger girls, but when asked to back up that claim, she pointedly does not mention she dated him*". This is in fact a direct quote from the scraped summaries, indicating that this is not an error by the agent nor the process of creating the graph database, but instead an error in the used data. In truth in the series, Spencer and Ian only share one kiss ("Spencer Hastings," n.d.).

4 Limitations

There were a few limitations to this experiment. The main limiting factor was most likely the usage of summaries instead of the original material, as a lot of information, especially that less relevant to the plot, was missing from the used summaries and therefore not included in any of the databases the model could extract context from. While this was fine for the rather broad test questions, for more detailed questions, more detailed data would be needed. Further the data is unreliable in its correctness as that the data was not released by official sources.

Secondly, another huge limiting factor laid in the usage of Large Language Models during various steps. During the process of creating the Graph Database, a LLM was implemented as an automatic extraction mechanism to collect entities and their respective relationships towards one another. That came with a few issues, due to the unreliability of such systems. The LLM did not always manage to strictly follow its given rules. For example, it invented new relationship types (e.g „ROMANTIC_INTEREST", "COUPLE" or "ACQUAINTANCE"). While these relationship types are added to the graph database, they cannot be retrieved later as the retrieval agent does only choose of the previously given relationship types. Further while the LLM did not invent new characters, it did sometimes invent surnames, when the surname was not originally given in the text. That had to be fixed during the data cleaning, but this is most likely a step that is rather hard to automize and therefore manual input is needed.

The LLM also failed to extract all characters and their relationships completely. This issue is particularly evident in the graph nodes: a notable number of nodes have no edges connecting them to other characters. For background characters, this may be plausible, as they often have few or no meaningful interactions with the main cast. However, the same problem occurs with more prominent characters. A clear example is Veronica Hastings in the series graph. Her node has no edges, even though in both adaptations she is

the mother of the main character Spencer Hastings and the wife of Peter Hastings—both of whom are present in the graph.

Lastly the implemented setup is also rather unflexible, especially in the realm of the Graph Data retrieval, as the LLM had to be constrained to a set of predefined functions, limiting the retrieval in its options. In this setup, relevant context can only be extracted if the question matches the available functions. Further, these functions also ignore relationships in the opposite direction, due to distinction between source and and target. It is possible that a more flexible approach would improve the results.

5 Conclusion & Learnings

All in all, the implemented system was rather successful at correctly answering the given questions and comparing the book implementations to the movie implementations. There was a slight tendency for misinformation (30%), but in both cases only parts of the data were incorrect. Further, in one of the cases the misbelief stemmed from incorrect given data, thus strictly going from the given data, only one of the six questions was answered incorrectly.

Interestingly enough, that question was one that could only be answered purely by the vector data context, due to the limited relationship types of the graph data. Hinting that the Graph Database might improve the output in comparison to a singular RAG system. Although to really deny or prove this theory, further research would be needed.

It should be once again noted in order to realize this approach, larger LLMs were needed. Smaller, locally run LLMs were often overwhelmed by the given tasks, not following the instructions and misunderstanding the input. While this approach seems to be very promising, the quality of its output is also very dependent on the resources available, as well as the data.

In future experiments the output could be improved through a few different methods:

- Using better, more general data. E.g. Using the whole book instead of summaries.
- Using more expensive embedding and reranking models to improve the relevancy of the extracted context from the vector data.
- Giving the graph retrieval more freedom, thus offering a higher diversity of the graph context.
- Using a larger model for entity and relationship extraction to reduce missing values.

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